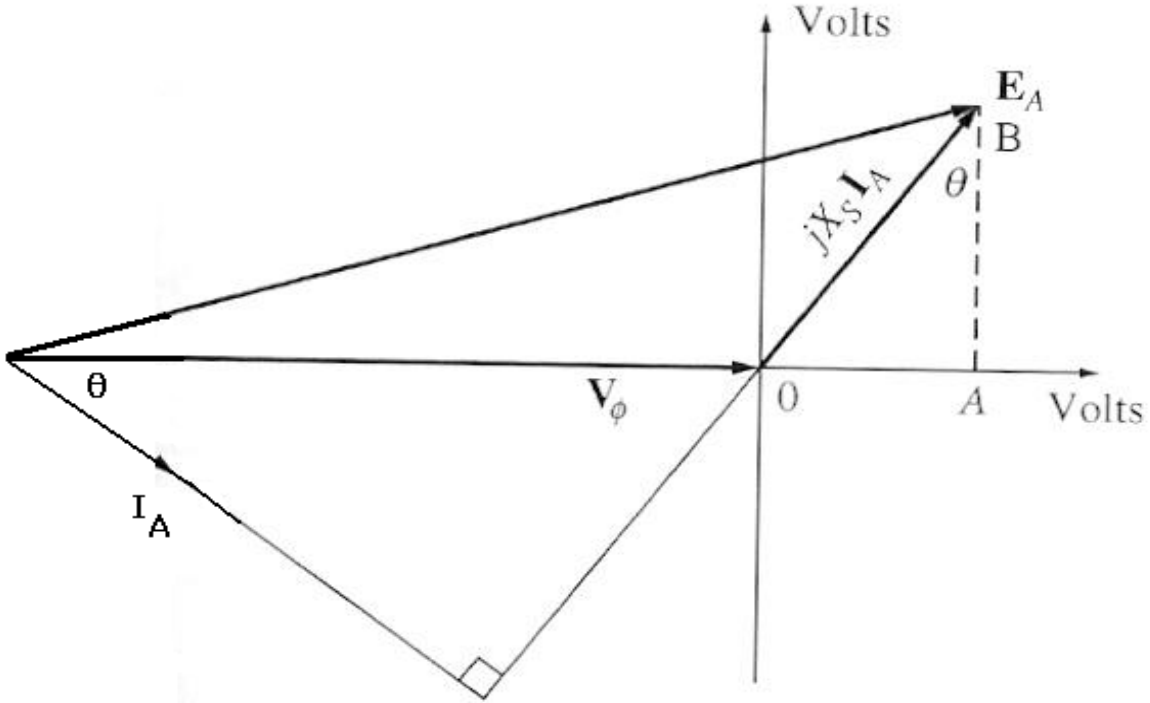
ENERGY CONVERSION ONE

(Course 25741)

CHAPTER FIVE
SYNCHRONOUS GENERATOR-CAPBILITY CURVE

SYNCHRONOUS GENERATOR

- A capability diagram is a plot of complex power S=P+jQ
- its curve can be derived back from voltage phasor diagram of the Syn. Gen.



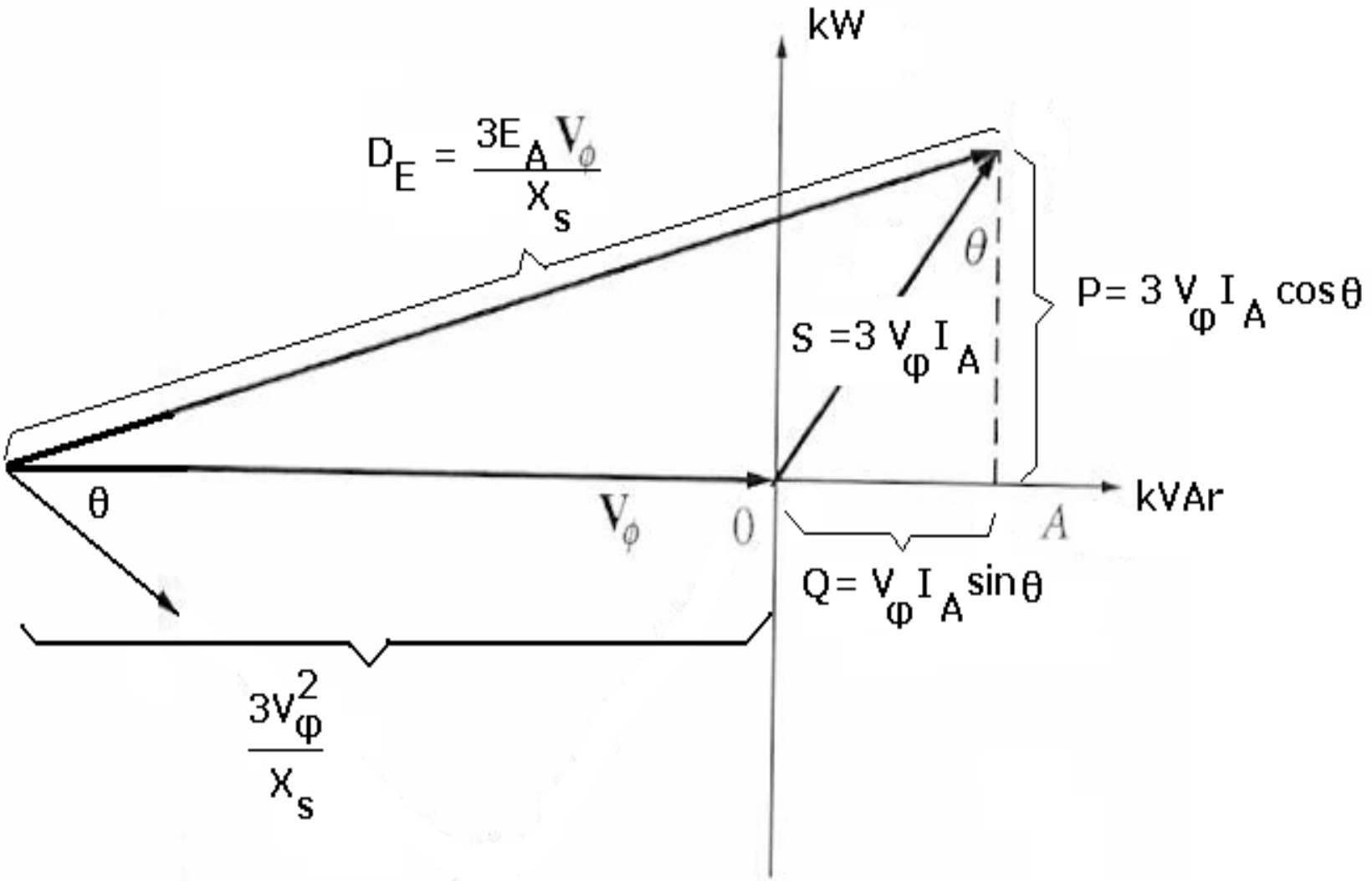
(a)

SYNCHRONOUS GENERATOR

- capability curve must represent power limits of generator, hence there is a need to convert the **voltage phasor** into **power phasor**.
- $P=3\ V_\varphi I_A \cos\theta$
- $Q=3\ V_\varphi I_A \sin\theta$
- $S= 3V_\varphi I_A$
- Reminding $P_{max}= 3\ V_\varphi E_A / X_s$ (5-21)
- The conversion factor to change scale of axes from V → VA is $3\ V_\varphi / X_s$

SYNCHRONOUS GENERATOR

- The corresponding power units

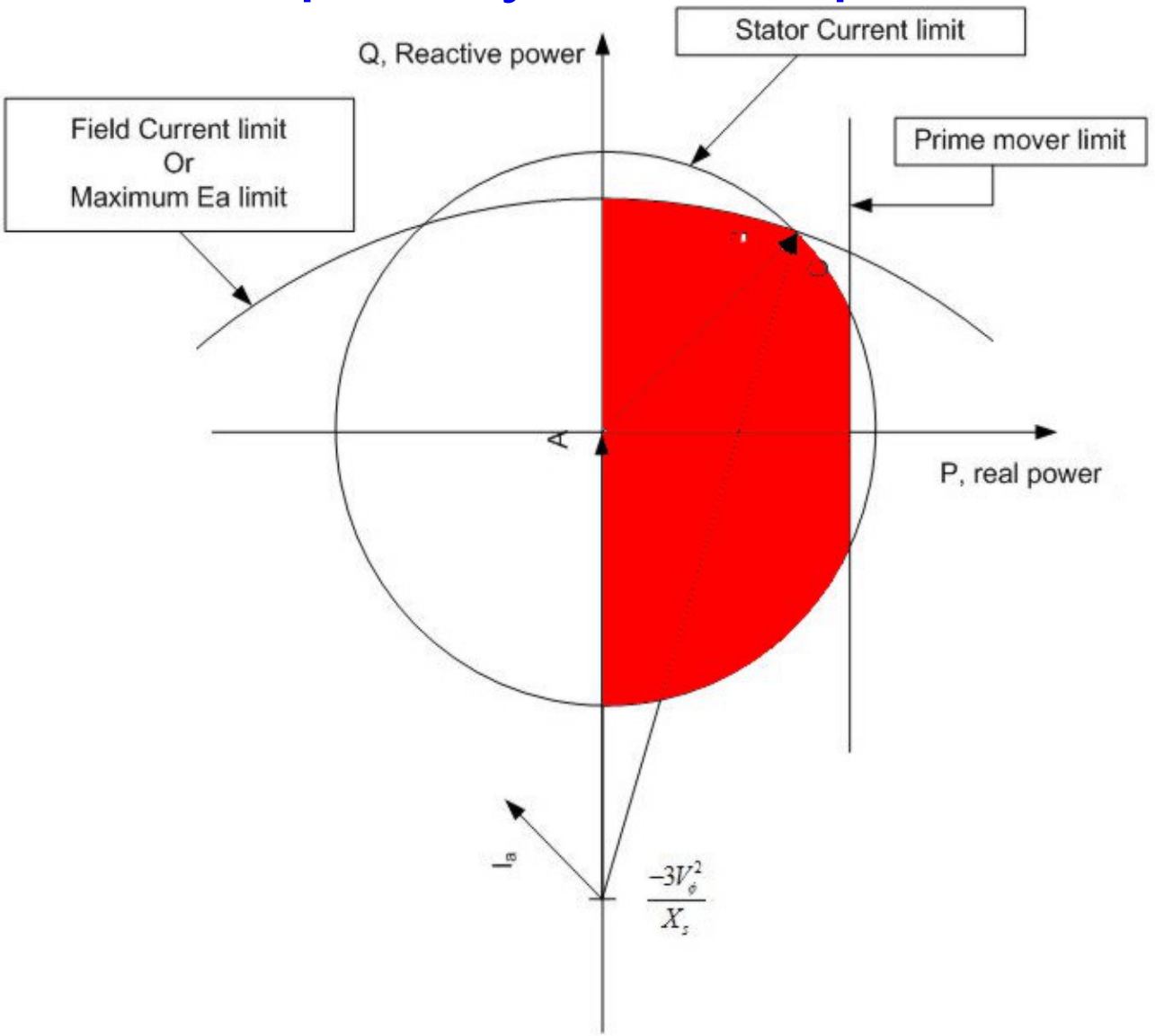


SYNCHRONOUS GENERATOR

- $P=3 V_\varphi I_A \cos\theta = 3 V_\varphi / X_s (X_s I_A \cos\theta)$
- $Q= 3 V_\varphi I_A \sin\theta = 3 V_\varphi / X_s (X_s I_A \sin\theta)$
- On voltage phasor diagram, origin of phasor diagram is at $-V_\varphi$ on horizontal axis, so origin on power diagram is:
- $Q = 3V_\varphi /X_s (-V_\varphi)=-3V_\varphi ^2/X_s$
- Field current ~ machine's flux & flux ~ $E_A=k\varphi\omega$
- Length corresponding to E_A on power diagram:
- $D_E=- 3 E_A V_\varphi / X_s$
- I_A ~ $X_s I_A$, and length corresponding to $X_s I_A$ on power diagram is $3 V_\varphi I_A$

SYNCHRONOUS GENERATOR

- Generator capability curve a plot of P versus Q



SYNCHRONOUS GENERATOR

- Any point lies within both circles is a safe operating point for generator
- However, the R.H.S. of Q axis means generator
- Also maximum prime-mover power & static stability limit should be considered

SYNCHRONOUS GENERATOR Capability Curve EXAMPLE

- A 480, 50 Hz, Y connected, six-pole syn. Gen. is rated at 50 kVA at 0.8 PF lagging. It has a synchronous reactance of 1.0 Ω per phase
  Assume generator connected to steam turbine capable of supplying up to 45 kW. The friction and windage losses are 1.5 kW, and core losses are 1.0 kW
- (a) sketch capability curve for this generator, including prime mover power limit
- (b)can this generator supply a line current of 56 A at 0.7 PF lagging? Why or why not
- (c) what is the maximum amount of reactive power this generator can produce
- (d) If generator supplies 30 kW of real power, what is maximum amount of reactive power that can be simultaneously supplied?

SYNCHRONOUS GENERATOR EXAMPLE-SOLUTION

- $S_{rated} = 3\, V_{\varphi,rated}\, I_{A,max}$
- $V_\varphi = V_T/\sqrt{3} = 480/\sqrt{3} = 277$ V
- $I_{A,max} = S_{rated}/3\, V_\varphi = 50$ kVA / (3x277)=60 A
- (a) maximum apparent power is 50 kVA, which specifies maximum safe armature current
- The center of E_A circle is at :
- $Q = -3\, V_\varphi\text{^}2/X_s = -3\ (277)\text{^}2 / 1.0 = -230$ kVAr
- Maximum of $E_A = V_\varphi + jX_s I_A = 277/_0 + (j1.0)(60/_-36.87) = 313 + j48 = 317\ /_8.7$ V
- Magnitude of distance ~ E_A is:
- $D_E = 3E_A V_\varphi / X_s = 3(317)(277)/1.0 = 263$ kVAr
- Maximum output power available with a prime-mover power of 45 kW is ≈
- $P_{max,out} = P_{max,in} - P_{mech\ loss} - P_{core\ loss} = 45 - 1.5 - 1.0 = 42.5$ kW

SYNCHRONOUS GENERATOR EXAMPLE-SOLUTION

(b) a current of 56 A at 0.7 PF lagging ➔

$P=3 V_{\varphi} I_A \cos\theta = 3x277x56x0.7=32.6$ kW

$Q=3 V_{\varphi} I_A \sin\theta = 3x277x56x0.714=33.2$ kVAr

Plotting this on capability diagram shows it is safely within maximum I_A curve, however outside maximum I_F curve, so this point is not a safe operating condition

(c) when real power supplied by Gen. zero, reactive power that generator can supply will be maximum. This point is right at peak of capability curve ➔ Q=263-230=33 kVAr

(d) if generator supplies 30 kW, maximum reactive power Gen. can supply 31.5 kVAr, the limiting factor is field current otherwise armature current is safe up to 39.8 kVAr